

of these MEAs has around hundred contact points that record the response of individual neurons as well as populations of neurons.

The microchips are first implanted in brain. The patient is made to think about doing something, like moving the hand forward or rolling the wrist. Each activity triggers a unique set of neurons. When a neuron is excited, a potential difference is developed between the inside and outside of the cell membrane and an ion current is developed. This ion current is converted into electron current by the sensors and sent to the decoder.

The difficult task is to interpret and associate the brain signals to their respective movements. With the help of revised filtering, processing and complex computer programs, and after many trials, the signals related to each activity can be figured out. The specially designed embedded software is programmed so as to receive each signal and make the connected external device (like a prosthetic hand) physically perform the activity associated with that particular signal.

Commercial viability

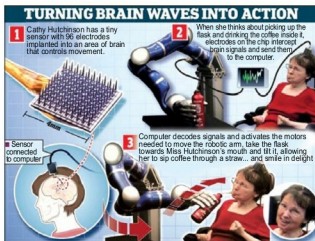
Though extensive researches are going on in this field, only a very small number of products are available in the consumer market; even lesser belonging to medical industry. High cost of production and unavailability of participants for clinical trials are a few reasons for their reduced popularity.

Some of the medical devices based on neurotechnology are discussed in brief below:

BrainGate implant. It is a brain implant developed by Cyberkinetics and is one of the most important BCI systems.

Deep brain stimulator. It is manufactured by Medtronic and is used to treat Parkinson's and dystonia. It is also researched for treatment of epilepsy and depression.

Hearing aid. The medical device company Otologics has made this product available in Europe.



Cathy Hutchinson (Courtesy: <http://www.dailymail.co.uk>)

Bionic eye. An artificial retina by Second Sight Medical Products, going through the first phase of trial.

Vagus nerve stimulator. It is used for treating epilepsy, depression and obesity. Cyberonics, MetaCure and EnteroMedics are the major manufacturers.

Powered exoskeleton. A robotic exoskeleton that can help the elderly and the disabled to walk and lift objects, developed by Cyberdyne.

Implanted defibrillator. This device is meant for people who have problems with irregular heartbeat. It prevents sudden cardiac arrest. It is developed by Medtronic, St. Jude Medical, Boston Scientific, Sorin Group CRM and Biotronic.

Ventricular assist. It helps the heart chamber pump blood. Thoratec, Abiomed, Inc. and World Health Corporation manufacture this product.

Breathing pacemaker. It helps people with spinal cord injury to help them breathe on their own by pacing diaphragm. The device is produced by Synapse Biomedical, Inc.

PillCam. A capsule with miniature camera that can be swallowed and is used for endoscopy. Given Imaging is the manufacturer of this capsule.

Drug delivery to spine. This drug delivery system is meant for patients with chronic pain. It is produced by Medtronic and MicroCHIPS.

Artificial arm. Prosthetic arms with 18 degrees of freedom, developed by

DEKA Research and Development Corporation.

Bionic hand. This product designed by Touch Bionics can perform more hand functions than a normal prosthetic hand.

Sacral nerve stimulation. Developed by Medtronic to control some bladder functions.

Smart leg. Most advanced lower limb prosthetic by Otto

Bock Healthcare.

Robotic foot. It has many added features over prosthetic foot. Ossur is the manufacturer of this product.

What future holds

As the next step of this technological advancement, researchers are planning to introduce feedback potentials to the electrodes, which can result in the interpretation of sensations like grip strength. They are also working on miniaturisation of equipment and introduction of wireless technology. It may even be possible to avoid the robotic devices by directly bypassing the appropriate signals to the corresponding motor nerves in the damaged section of the spinal cord. This means the paralysed patient will be able to move his/her own body.

The field of BMI has made a considerable development in the past decade. Though only few products are available in the market, extensive research and lab works are going on in this field. What was once just an interesting science fiction theme is now growing very close to reality. Neurotechnology has developed from just a figment of imagination to monkeys moving cursors on computer screen to humans controlling prosthetic limbs in three-dimensional space with dexterity. In coming years, we can expect BCI to not only restore human functions but also augment it and improve the quality of living. ●

The author is a technical correspondent at EYF